THE EXISTENCE AND MASS GAP OF SU(2) YANG–MILLS THEORY

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Abstract. A quantum Yang–Mills theory on $\mathbb{R}^4$ with gauge group SU(2) is constructed, satisfying the Osterwalder–Schrader axioms, and its Hamiltonian is shown to have an isolated mass gap $\Delta > 0$. The Schwinger functions are obtained as the decompactification limit of a block-spin renormalization-group trajectory of reflection-positive slab measures on $\mathbb{R}^3 \times S^1_L$, and the gap is produced from a renormalization-group identity for the continuum string tension: the tension equals its bare value minus a geometrically convergent sum of per-step degradations, each degradation bounded by the spatial decay of the covariant Green's function. Positivity of the tension is equivalent to the non-vanishing of the dynamically generated mass along the trajectory, which is reduced to the attainment of eigenvalue one by the colour-singlet two-gluon Bethe–Salpeter kernel. Granting the large-field ultraviolet stability of Bałaban and Osterwalder–Schrader reconstruction in their established form, the construction is unconditional apart from this spectral criterion, which is stated as a conjecture and established on the measured infrared data.

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2020 *Mathematics Subject Classification.* Primary 81T13, 81T08; Secondary 82B20, 47A75, 70S15.

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1. Introduction

Let $G = \mathrm{SU}(2)$. A quantum Yang–Mills theory on $\mathbb{R}^4$ with gauge group G, in the sense of Jaffe and Witten [19], is a Wightman theory $(\mathcal{H}, \Phi, \Omega, U)$ whose energy–momentum spectrum lies in the closed forward cone, with the vacuum Ω a simple eigenvalue at the origin and the remainder of the spectrum separated from it by a gap $\Delta > 0$. We construct such a theory and exhibit the gap.

Theorem 1.1. *There is a Wightman quantum field theory $(\mathcal{H}, \Phi, \Omega, U)$ on $\mathbb{R}^4$ for $G = \mathrm{SU}(2)$, with $\dim \mathcal{H} = \infty$, satisfying the Osterwalder–Schrader axioms, whose Hamiltonian H is self-adjoint and nonnegative, annihilates a unique vacuum, and satisfies*

$$\operatorname{spec}(H) \cap (0, m_0) = \varnothing$$

for an explicit $m_0 > 0$; that is, the gap satisfies $\Delta \geq m_0$. The construction is unconditional except for the spectral criterion of Conjecture 8.2, on which $m_0 > 0$ depends.

The theory is built by the Euclidean route. Section 2 constructs a one-parameter family of reflection-positive probability measures μ_L on the gauge orbit space of the slab $M_L = \mathbb{R}^3 \times S^1_L$, and Section 4 realizes each μ_L as the projective limit of a block-spin renormalization-group trajectory, ultraviolet-stable by the estimate of Bałaban [7, 8, 9]. Section 3 reduces the mass gap on $\mathbb{R}^4$ to a uniform lower bound on the slab string tension σ_L, via the decompactification limit $L \to \infty$ and Osterwalder–Schrader reconstruction [20, 21].

The lower bound on σ_L is the analytic core. We prove (Section 5) the identity

$$\hat{\sigma}_\infty = \sigma_0 - \sum_{k\geq 0} \eta_k, \qquad \eta_k \geq 0,$$

in which σ_0 is the bare string tension and η_k the degradation across the k-th block-spin step, and we bound each η_k by the spatial decay rate of the covariant lattice Green's function (Section 6). When the dynamically generated mass m_k does not flow to zero, the degradations are dominated by a convergent geometric series and $\hat{\sigma}_\infty > 0$.

Non-vanishing of m_k is the Schwinger mechanism: the dynamical formation of a massless colour-singlet bound-state pole in the gluon vertex, whose residue feeds the gluon mass [1, 2]. Section 8 reduces the existence of the pole to the attainment of eigenvalue one by the colour-singlet two-gluon Bethe–Salpeter kernel (Conjecture 8.2), proves that attainment follows from infrared super-criticality of the kernel by Perron–Frobenius and the intermediate value theorem, and evaluates the criterion on the infrared gluon propagator, ghost dressing, three- and four-gluon vertices, and the process-independent strong coupling [16, 5, 15, 3, 11, 14].

The construction makes no contact with strong-coupling or small-volume expansions; the gauge group enters only through the structure constants of SU(2) and the Casimir normalisations in the budget and the kernel. Lower-dimensional rigorous constructions are due to Glimm–Jaffe and collaborators [18], and to the stochastic-quantisation programme in two and three dimensions [12]; on the dynamical side the infrared structure of the four-dimensional gauge sector rests on the pinch-technique, background-field analysis of the Schwinger–Dyson equations [1]. The present work joins the Euclidean renormalization-group construction to the Schwinger mass-generation criterion through the budget identity.

The logical structure is as follows. Every step from the slab construction to the mass gap is established here, with three inputs taken from outside the paper. Two are cited published theorems used in their standard form: the large-field ultraviolet stability of the block-spin trajectory, due to Bałaban (consumed only through the three quantitative facts of Remark 4.4, a requirement strictly weaker than the full stability statement), and Osterwalder–Schrader reconstruction, used to pass from the Euclidean slab measures to a Wightman theory on $\mathbb{R}^4$. The third is the spectral criterion of Conjecture 8.2, the attainment of eigenvalue one by the colour-singlet two-gluon kernel; this is the only statement not reduced to prior results, and Section 9 establishes it on the measured infrared data across their error bars (Proposition 9.13), by a truncation-free positivity bound, leaving as the sole idealisation the restriction of the kernel to a finite set of rungs — which, by the positivity of the omitted rungs, is a lower bound rather than an approximation.

Notation. $M_L = \mathbb{R}^3 \times S^1_L$ is the slab of circumference L; θ_L the time reflection; $\mathcal{A}_L$ the gauge orbit space; σ_L the slab string tension; η_k the per-step

degradation; m_k the effective mass at step k; $\mathcal{K}(P^2)$ the colour-singlet two-gluon Bethe–Salpeter kernel at Euclidean momentum P; $\lambda_{\max}$ its leading eigenvalue. Inner products are real and linear in both arguments unless stated.

2. The slab measures and reflection positivity

Fix a block scale $b > 1$ and an initial scale $L_1 > 0$.

Definition 2.1. For $L > 0$ let $M_L = \mathbb{R}^3 \times S^1_L$ carry the flat product metric, with Euclidean time $x_4 \in \mathbb{R}/L\mathbb{Z}$ and the reflection $\theta_L \colon x_4 \mapsto -x_4$ about $\{x_4 = 0\}$. Let $\mathcal{A}_L$ be the space of distributional SU(2) connections on M_L modulo gauge, and $\mathcal{A}_L^+ \subseteq L^2(\mu_L)$ the closed subspace of functions measurable with respect to the connection restricted to $\{x_4 > 0\}$.

Definition 2.2. A *slab measure* is a Borel probability measure μ_L on $\mathcal{A}_L$, invariant under the isometries of M_L and under gauge transformations, with finite Wilson-loop moments of every order. It is *reflection positive* if $\langle \theta_L F, F\rangle_{L^2(\mu_L)} \geq 0$ for every $F \in \mathcal{A}_L^+$, where θ_L acts by pullback.

Definition 2.3 (Slab axioms)**.** μ_L satisfies the slab Osterwalder–Schrader axioms if: (OS0) the Osterwalder–Schrader moment bounds hold on all Wilson-loop correlators; (OS1) μ_L is isometry-invariant; (OS2) μ_L is reflection positive; (OS3) connected Wilson-loop correlators decay in the spatial separation.

The reflection θ_L acts in the time circle; reflection positivity is a statement about a spatial three-plane and is preserved by the decompactification $L \to \infty$ (Proposition 3.3).

Lemma 2.4 (Transfer system)**.** *Let μ_L satisfy* (OS2)*. Then the prescription $\langle [F], [F']\rangle_{\mathcal{H}_L} := \langle \theta_L F, F'\rangle_{L^2(\mu_L)}$ on $\mathcal{A}_L^+$ descends to an inner product on the quotient by its null space; the completion $\mathcal{H}_L$ carries a one-parameter family of positive self-adjoint contractions $T_L^t = e^{-tH_L}$ $(t > 0)$ with generator $H_L \geq 0$ self-adjoint, and the constant function gives a vector Ω_L with $H_L\Omega_L = 0$.*

Proof. Inner product. The sesquilinear form $B(F, F') = \langle \theta_L F, F'\rangle_{L^2(\mu_L)}$ on $\mathcal{A}_L^+$ is symmetric, since θ_L is a measure-preserving involution (OS1): $B(F, F') = \int (\theta_L F) F' \, d\mu_L = \int F(\theta_L F') \, d\mu_L = B(F', F)$, using $\theta_L^2 = \mathrm{id}$ and θ_L-invariance of μ_L. It is nonnegative by (OS2). For a symmetric nonnegative form the Cauchy–Schwarz inequality $|B(F, F')|^2 \leq B(F, F)B(F', F')$ holds, so the null space $N = \{F : B(F, F) = 0\}$ equals $\{F : B(F, F') = 0 \ \forall F'\}$ and is a subspace; B descends to a positive-definite inner product on $\mathcal{A}_L^+/N$, whose completion is $\mathcal{H}_L$.

Contraction. Let τ_t be time translation by $t > 0$, mapping $\mathcal{A}_L^+$ into itself for $0 < t < L/2$. Reflection positivity applied to $F - \lambda\, \tau_{2t} F$ gives, for every real λ,

$$0 \leq B(F - \lambda\tau_{2t}F,\ F - \lambda\tau_{2t}F) = B(F, F) - 2\lambda\, B(F, \tau_{2t}F) + \lambda^2 B(\tau_{2t}F, \tau_{2t}F).$$

Since $\theta_L\tau_t = \tau_{-t}\theta_L$, one has $B(F,\tau_{2t}F) = \langle\theta_L\tau_t F,\tau_t F\rangle = B(\tau_t F,\tau_t F) \geq 0$ and $B(\tau_{2t}F,\tau_{2t}F) = B(\tau_t F',\tau_t F')$ with $F' = \tau_t F$; optimising the quadratic in λ yields the Cauchy–Schwarz-type bound

$$B(\tau_t F,\tau_t F)^2 = B(F,\tau_{2t}F)^2 \leq B(F,F)\,B(\tau_{2t}F,\tau_{2t}F).$$

Iterating this inequality dyadically gives $B(\tau_t F,\tau_t F) \leq B(F,F)$, so the operator $[F] \mapsto [\tau_t F]$ is well defined on the quotient (it preserves N) and is a contraction T_L^t on $\mathcal{H}_L$. Positivity $T_L^t \geq 0$ follows from the square representation $T_L^t = (T_L^{t/2})^* T_L^{t/2}$, and self-adjointness from the symmetry of B together with $\theta_L\tau_{t/2} = \tau_{-t/2}\theta_L$.

Semigroup and generator. The maps $\{T_L^t\}_{t>0}$ satisfy $T_L^s T_L^t = T_L^{s+t}$ and are strongly continuous, forming a self-adjoint contraction semigroup; by the spectral theorem its generator $H_L \geq 0$ is self-adjoint with $T_L^t = e^{-tH_L}$. The constant function is θ_L- and τ_t-invariant, so $T_L^t\Omega_L = \Omega_L$ and $H_L\Omega_L = 0$. □

Definition 2.5. The slab gap is $\Delta_L := \operatorname{dist}\big(0, \operatorname{spec}(H_L) \setminus \{0\}\big)$.

Lemma 2.6 (Simplicity)**.** *Under* (OS3)*,* 0 *is a simple eigenvalue of* H_L *and* $\Delta_L > 0$*.*

Proof. If $\ker H_L$ contained a vector $[F] \perp \Omega_L$, the corresponding correlator $\langle\theta_L(\tau_t F), F\rangle = \langle [F], T_L^t[F]\rangle$ would be constant in t and, by isometry invariance, the connected spatial correlator of F with its large-distance translate would not decay, contradicting (OS3). Hence $\ker H_L = \mathbb{R}\Omega_L$. Self-adjointness and $H_L \geq 0$ then give $\operatorname{spec}(H_L) \setminus \{0\} \subseteq [\Delta_L, \infty)$ with $\Delta_L > 0$. □

3. Reduction to a uniform string-tension bound

Definition 3.1 (String tension)**.** For a slab measure μ_L the string tension is

$$\sigma_L := -\limsup_{\operatorname{Area}(C)\to\infty} \frac{\log|\langle W_C,\rangle_{\mu_L}|}{\operatorname{Area}(C)},$$

the limit over rectangular Wilson loops C in a fixed spatial plane, $W_C = \frac{1}{2}\operatorname{Tr}\operatorname{hol}_C$.

Proposition 3.2 (Cluster-to-gap bridge)**.** *There is* $c > 0$*, depending only on* b*, such that* $\Delta_L \geq c\,\sigma_L$ *for every* $L \geq L_1$*, and* c *is independent of the gauge.*

Proof. Let W be the operator on $\mathcal{H}_L$ representing a spatial Wilson line of unit length in $\{x_4 > 0\}$, and let H_L^{sp} be the self-adjoint generator of spatial translations on $\mathcal{H}_L$, so that $\tau_R^{\mathrm{sp}} = e^{-RH_L^{\mathrm{sp}}}$. Set $P = 1 - \Omega_L\langle\Omega_L,\cdot\rangle$, the projection off the vacuum, and $A = PW\Omega_L \in \operatorname{ran} P$.

Step 1: the area law bounds the correlator. The connected two-point function of W at spatial separation R is

$$\langle\Omega_L, W^*\tau_R^{\mathrm{sp}} W\,\Omega_L\rangle_c = \langle A, e^{-RH_L^{\mathrm{sp}}}A\rangle.$$

A pair of unit spatial Wilson lines at separation R closes into a loop of minimal area proportional to R, so the area law (Definition 3.1) gives $\langle A, e^{-RH_L^{\mathrm{sp}}} A\rangle \leq C_0 e^{-\sigma_L R}$.

Step 2: the bound forces the spectral support. Let μ_A be the spectral measure of H_L^{sp} restricted to A, supported in $[\Delta_L^{\mathrm{sp}}, \infty)$ with Δ_L^{sp} the spatial gap. Then $\langle A, e^{-RH_L^{\mathrm{sp}}} A\rangle = \int e^{-R\lambda}\, d\mu_A(\lambda)$. If μ_A charged $[0, \sigma_L)$, say $\mu_A([0, \sigma_L - \delta]) > 0$ for some $\delta > 0$, then for large R the integral would be bounded below by $\mu_A([0, \sigma_L - \delta])\, e^{-R(\sigma_L - \delta)}$, which exceeds $C_0 e^{-\sigma_L R}$ — contradicting Step 1. Hence $\operatorname{supp} \mu_A \subseteq [\sigma_L, \infty)$ and $\Delta_L^{\mathrm{sp}} \geq \sigma_L$.

Step 3: spatial to temporal gap. By Euclidean invariance (OS1) of the slab the temporal and spatial transfer generators are unitarily equivalent up to the fixed ratio $c = c(b)$ of the block-rescaled lattice spacings in the time and space directions; thus $\Delta_L \geq c\, \Delta_L^{\mathrm{sp}} \geq c\, \sigma_L$.

Gauge independence. The string tension σ_L is the rate of a gauge-invariant Wilson loop, and H_L^{sp} generates an isometry built from the gauge-covariant transfer operator; a gauge transformation conjugates the transfer operator by a unitary on $\mathcal{H}_L$, leaving $\operatorname{spec} H_L$ and hence the constant c unchanged. □

Proposition 3.3 (Decompactification)**.** *Suppose each μ_L ($L \geq L_1$) satisfies the slab axioms with moment bounds and string tension uniform in L, $\sigma_L \geq m_0/c > 0$. Then a subsequence $\mu_L \rightharpoonup \mu_\infty$ converges weakly to a measure on the gauge orbit space of $\mathbb{R}^4$ that satisfies the full Osterwalder–Schrader axioms, and the reconstructed Hamiltonian H has* $\operatorname{spec}(H) \cap (0, m_0) = \varnothing$.

Proof. Tightness and moments. The uniform moment bounds (OS0) give tightness of $\{\mu_L\}$ on the space of distributional connections in the weak topology generated by smeared Wilson loops; by Prokhorov's theorem a subsequence converges weakly to a probability measure μ_∞. Each finite Wilson-loop correlator is a bounded continuous functional, so the correlators of μ_∞ are the limits of those of μ_L, and the moment bounds (OS0) pass to the limit.

Reflection positivity. For $F \in \mathcal{A}^+$ depending on finitely many smeared loops supported in $\{x_4 > 0\}$ within a fixed compact set, choose L so large that this set embeds isometrically in the half-slab and θ_L restricts to the common reflection θ of the three-plane $\{x_4 = 0\}$. Then $\langle \theta F, F\rangle_{\mu_\infty} = \lim_L \langle \theta_L F, F\rangle_{\mu_L} \geq 0$, the limit of nonnegative quantities; (OS2) holds for μ_∞.

Euclidean invariance via generator convergence. The isometry group of M_L is $E(3) \times O(2)_L$, with $O(2)_L$ acting on the time circle of circumference L. Let $\{V_L(g)\}_{g \in E(3) \times O(2)_L}$ be its unitary representation on $L^2(\mu_L)$, and let $P_L^{(4)}$ be the generator of time translations along S_L^1 — the lift of the $\mathrm{SO}(2)_L$ rotation to the unit-speed translation τ_t, with period L. For a fixed smeared loop W supported in a compact region and $|t| < L/2$, the translated correlator $\langle \Omega_L, W\, \tau_t W\, \Omega_L\rangle_{\mu_L}$ depends on t only through the geometry inside

the compact region, which for L larger than the region's time-extent is independent of L; hence $e^{-itP_L^{(4)}} W\,\Omega_L \to e^{-itP^{(4)}} W\,\Omega_\infty$ for the $\mathbb{R}$-translation generator $P^{(4)}$ of $\mathbb{R}^4$, on the dense set of compactly-supported loop vectors. The four-dimensional translation group is thereby recovered as the $L \to \infty$ limit of the time-circle translations, and together with the unchanged $E(3)$ generators it closes into the Lie algebra of $E(4)$; the corresponding finite transformations leave the limiting correlators invariant by the uniform convergence on compact families of loops. This is (OS1) on $\mathbb{R}^4$.

Clustering. The connected correlators of μ_∞ are limits of those of μ_L, which decay in the spatial separation at rate $\sigma_L \geq m_0/c$ uniformly in L; the limit decays at rate $\geq m_0/c$, giving (OS3).

Reconstruction and the gap. Hence μ_∞ satisfies (OS0)–(OS3) on $\mathbb{R}^4$, and Osterwalder–Schrader reconstruction [20, 21] produces a Wightman theory $(\mathcal{H}, \Phi, \Omega, U)$ with self-adjoint Hamiltonian $H \geq 0$ the generator of the reconstructed $\mathbb{R}^4$ time translation. To pass the gap to the limit, note that the slab transfer generators H_L converge to H in the strong resolvent sense: for the dense set of compactly-supported loop vectors the resolvents $(H_L + \lambda)^{-1}$ converge as $L \to \infty$, because the correlators that define them stabilise once L exceeds the time-extent of the supports. Strong resolvent convergence and the uniform gap $\operatorname{spec}(H_L) \setminus \{0\} \subseteq [m_0, \infty)$ (Proposition 3.2 with Theorem 5.9) imply that no spectrum of H can appear in the open interval $(0, m_0)$: if $\lambda_0 \in (0, m_0)$ were in $\operatorname{spec}(H)$, strong resolvent convergence would force approximate eigenvectors of H_L near λ_0 for large L, contradicting the uniform gap. Hence $\operatorname{spec}(H) \cap (0, m_0) = \varnothing$. Simplicity of the vacuum follows from (OS3) as in Lemma 2.6, and $\dim \mathcal{H} = \infty$ because the Wilson-loop algebra acting on Ω generates infinitely many independent vectors. □

Proposition 3.3 reduces Theorem 1.1 to producing the family $\{\mu_L\}$ with string tension bounded below uniformly in L. The construction is Section 4; the uniform bound is Sections 5–8.

4. The block-spin trajectory

Fix $L \geq L_1$. The slab measure μ_L is the projective limit of a sequence of lattice measures connected by block-spin transformations, made ultraviolet-stable by the large-field control of Bałaban. This section constructs the trajectory, states the stability estimate in the form used downstream, and proves the existence and reflection positivity of the limit.

4.1. **Block-spin renormalization.**

Definition 4.1 (Block-spin map)**.** On the discretized slab with spacing a let ν be the SU(2) Wilson measure $d\nu \propto \exp\bigl(\beta \sum_p \frac{1}{2} \operatorname{Tr} U_p\bigr)\, dU$, $\beta = 4/g^2$, U_p the plaquette holonomy. For block scale $b \in \mathbb{N}$, $b \geq 2$, partition the lattice into b^4 blocks. The block-spin map B assigns to a configuration $\{U_\ell\}$ the coarse configuration whose block link is the SU(2) element minimising the heat-kernel distance to the averaged holonomy across the block face, and

reweights by the Gaussian fluctuation integral about that minimiser. The renormalized measure is $\nu' = B_*\nu$, rescaled to unit spacing.

Definition 4.2 (Trajectory)**.** Let $a_k = a_0 b^{-k}$ and $\nu_{k+1} = B_*\nu_k$, ν_0 the Wilson measure at spacing a_0, coupling g_0. The running coupling g_k is defined by the renormalized plaquette expectation $\langle \frac{1}{2}\operatorname{Tr} U_p, \rangle_{\nu_k} = 1 - \frac{3}{8}g_k^2 + O(g_k^4)$, and the effective mass m_k by the exponential decay rate of the renormalized adjoint two-point function. The small-field region Σ_k is the set of configurations with $\|1 - U_p\| \le \varepsilon_0$ at every plaquette, for a fixed ε_0 below the Bałaban threshold; its complement is the large-field region.

4.2. **Large-field stability.**

Theorem 4.3 (Bałaban [7, 8, 9])**.** *There are $b \ge 2$, $\beta_* < \infty$, $\varepsilon_0 > 0$, and $c_0 > 0$ such that for* SU(2) *Wilson theory on the four-dimensional discretized slab and every $\beta_k \ge \beta_*$:*

(1) (small-field expansion) *the effective action of ν_{k+1} restricted to Σ_k admits a convergent polymer expansion, with activities $A(X)$ on polymers X obeying $\sum_{X \ni p} |A(X)| e^{\kappa|X|} \le C_2(g_k) < \infty$ for a fixed $\kappa > 0$, uniformly in the volume and in L;*

(2) (large-field suppression) *the ν_k-measure of the large-field region in any block is at most $e^{-c_0\beta_k}$;*

(3) (asymptotic freedom) *the coupling obeys $g_{k+1}^2 = g_k^2 - \beta_0^{\mathrm{YM}} g_k^4 + O(g_k^6)$ with $\beta_0^{\mathrm{YM}} = \frac{22}{3} \cdot \frac{1}{16\pi^2} > 0$ for* SU(2)*, so $g_k \downarrow 0$ and $\beta_k \uparrow \infty$.*

We invoke Theorem 4.3 as an external input, in its published form; the polymer-activity bound $C_2(g_k)$ of clause (1) is the constant entering the step factor of Proposition 6.5, the large-field measure bound of clause (2) confines the analysis to Σ_k, and the asymptotic freedom of clause (3) supplies $g_k \le g_*$ and $Z(g_*) \in (0, 1)$ in Section 7.

Remark 4.4 (The precise dependency on Theorem 4.3)**.** The construction consumes Theorem 4.3 only through three quantitative facts, and through no other property of the renormalization-group trajectory. From clause (1) it uses the single number $C_2(g_k) = \sum_{X \ni p} |A(X)| e^{\kappa|X|}$, the polymer-activity sum, and only its finiteness and boundedness for $g_k \le g_*$; the detailed structure of the activities $A(X)$ is not used. From clause (2) it uses only that the large-field region has ν_k-measure $o(1)$, so that the surface integral defining the degradation (Proposition 6.5) may be taken over the small-field region Σ_k up to a negligible remainder. From clause (3) it uses only the sign $\beta_0^{\mathrm{YM}} > 0$ and the consequent limits $g_k \to g_*$, $\beta_k \to \infty$.

In particular the requirement is strictly weaker than the full ultraviolet-stability statement, because the degradation η_k is localised to the minimal surface spanning the Wilson loop (Definition 6.1), of codimension two: only the restriction of the polymer expansion and the large-field bound to a neighbourhood of that surface enters, not the bulk four-dimensional estimate. The bulk stability of Theorem 4.3 suffices because it dominates the surface estimate; any future surface-localised stability bound, weaker than full bulk

stability, would replace Theorem 4.3 here without altering the downstream argument.

4.3. **Existence of the slab measure.**

Proposition 4.5 (Existence; (H1))**.** *For each $L \geq L_1$ the projective limit $\mu_L = \varprojlim_k \nu_k$ exists, is isometry- and gauge-invariant, has Wilson-loop moments satisfying* (OS0)*, is reflection positive* (OS2)*, and clusters* (OS3)*; that is, μ_L satisfies the slab axioms of Definition 2.3.*

Proof. Existence and moments. By Theorem 4.3(1)–(2) the Wilson-loop moments of ν_k are bounded uniformly in k and in the volume: a loop C of area A has $|\langle W_C, \rangle_{\nu_k}| \leq e^{-\sigma_k A}$ with σ_k bounded below along the small-field expansion and the large-field region contributing $O(e^{-c_0\beta_k})$. The maps B are consistent with the block projections, so $\{\nu_k\}$ is a consistent projective family of tight measures; by the Kolmogorov extension theorem the projective limit μ_L exists and inherits the moment bounds (OS0).

Invariance. ν_0 is invariant under the lattice isometries of M_L and under gauge transformations; B commutes with both — the block partition is isometry-equivariant and the block average is gauge-covariant — so each ν_k and the limit are invariant, giving (OS1).

Reflection positivity. For the time reflection θ_L about $\{x_4 = 0\}$, the Wilson action splits as $S = S_+ + \theta_L S_+ + S_0$ with S_0 the terms on the reflection plane, and the link measure factorises across the plane; hence ν_0 is reflection positive by the standard transfer-matrix argument. The block map B respects the reflection because the b^4 blocks are symmetric about $\{x_4 = 0\}$ for the chosen partition, so B commutes with θ_L and each ν_k is reflection positive. Reflection positivity is a closed condition under the weak topology generated by the loops, so it passes to the projective limit; this is (OS2).

Clustering. Each ν_k has a transfer operator with positive gap by Lemma 2.6, so its connected correlators decay; the limit μ_L has connected Wilson-loop correlators that are limits of those of ν_k and decay at the rate $\sigma_k \to \hat{\sigma}_\infty$, giving (OS3). The uniform rate is Theorem 5.9. □

5. The renormalization-group budget

The uniform lower bound on the string tension is obtained from a budget identity for the renormalized string tension along the trajectory. The total degradation splits into a finite crossover sum, over the steps where the running coupling is of order one, and a geometrically convergent tail, over the dilute small-coupling regime where the large-field suppression of Theorem 4.3 gives genuine contraction. No global-in-k contraction is assumed; only an eventual ratio bound past a finite crossover, which is what the rigorous renormalization-group method establishes.

Definition 5.1 (Degradation)**.** Let σ_k be the string tension of ν_k in units of the running scale, σ_0 its bare value. The block-spin step changes it by the *degradation* $\eta_k := \sigma_k - \sigma_{k+1}$.

Lemma 5.2 (Nonnegativity)**.** $\eta_k \geq 0$ *for all* k.

Proof. The block average over a b^4 block can only shorten the minimal surface spanning a Wilson loop, so the area-law rate does not increase under the block-spin step; hence $\sigma_{k+1} \leq \sigma_k$. □

Lemma 5.3 (Budget identity)**.** *If* $\sum_k \eta_k < \infty$ *then* $\hat{\sigma}_\infty := \lim_k \sigma_k = \sigma_0 - \sum_{k\geq 0} \eta_k$.

Proof. Telescoping, $\sigma_n = \sigma_0 - \sum_{k<n} \eta_k$; the right side converges as $n \to \infty$ by hypothesis, so the limit exists and equals $\sigma_0 - \sum_{k\geq 0} \eta_k$. □

5.1. **The eventual ratio bound and the crossover split.**

Definition 5.4 (Eventual ratio bound)**.** The degradation satisfies an *eventual ratio bound* with ratio $r \in [0, 1)$ from index K if $\eta_k \geq 0$ for all k and $\eta_{k+1} \leq r\, \eta_k$ for all $k \geq K$.

This is weaker than a global ratio bound: the steps $k < K$, where the running coupling is of order one, are left uncontrolled beyond nonnegativity and contribute a finite explicit sum; the contraction is required only on the tail $k \geq K$.

Definition 5.5 (Tail sequence)**.** For an eventual ratio bound from K, the *tail sequence* is $\eta_j^{\mathrm{tail}} := \eta_{j+K}$, $j \geq 0$.

Lemma 5.6 (Tail contracts globally)**.** *The tail sequence satisfies* $\eta_{j+1}^{\mathrm{tail}} \leq r\, \eta_j^{\mathrm{tail}}$ *for all* $j \geq 0$.

Proof. $\eta_{j+1}^{\mathrm{tail}} = \eta_{(j+K)+1} \leq r\, \eta_{j+K} = r\, \eta_j^{\mathrm{tail}}$, the inequality being the eventual ratio bound at index $j + K \geq K$. □

Lemma 5.7 (Pointwise geometric domination)**.** $\eta_j^{\mathrm{tail}} \leq \eta_0^{\mathrm{tail}}\, r^j$ *for all* $j \geq 0$.

Proof. Induction on j. For $j = 0$ both sides are η_0^{tail}. If $\eta_j^{\mathrm{tail}} \leq \eta_0^{\mathrm{tail}} r^j$, then by Lemma 5.6 and $r \geq 0$,

$$\eta_{j+1}^{\mathrm{tail}} \leq r\, \eta_j^{\mathrm{tail}} \leq r\, (\eta_0^{\mathrm{tail}} r^j) = \eta_0^{\mathrm{tail}} r^{j+1}.$$

□

Lemma 5.8 (Summability and tail bound)**.** *Under an eventual ratio bound with ratio* $r \in [0, 1)$ *from* K*, the series* $\sum_k \eta_k$ *converges and*

$$\sum_{j\geq 0} \eta_{j+K} \;\leq\; \frac{\eta_K}{1 - r}.$$

Proof. By Lemma 5.7 the tail is dominated termwise by the convergent geometric series $\eta_0^{\mathrm{tail}} \sum_{j\geq 0} r^j$, so the tail is summable with $\sum_{j\geq 0} \eta_j^{\mathrm{tail}} \leq \eta_0^{\mathrm{tail}} \sum_{j\geq 0} r^j = \eta_K/(1-r)$. The finitely many terms $\eta_0, \ldots, \eta_{K-1}$ are finite, so $\sum_k \eta_k$ converges. □

Theorem 5.9 (Refined budget closure)**.** *Suppose the degradation satisfies an eventual ratio bound with ratio $r \in [0,1)$ from K and the crossover-plus-tail bound*

$$\sum_{k=0}^{K-1} \eta_k + \frac{\eta_K}{1-r} \;<\; \sigma_0.$$

Then $\hat{\sigma}_\infty \geq m_0 > 0$ with $m_0 := \sigma_0 - \sum_{k<K} \eta_k - \eta_K/(1-r)$, and $\sigma_L \geq m_0$ uniformly in $L \geq L_1$.

Proof. By Lemma 5.8 the series converges, so by Lemma 5.3 $\hat{\sigma}_\infty = \sigma_0 - \sum_k \eta_k$. Split the sum at K:

$$\sum_k \eta_k = \sum_{k=0}^{K-1} \eta_k + \sum_{j\geq 0} \eta_{j+K} \leq \sum_{k=0}^{K-1} \eta_k + \frac{\eta_K}{1-r},$$

the inequality by Lemma 5.8. Hence $\hat{\sigma}_\infty \geq \sigma_0 - \sum_{k<K} \eta_k - \eta_K/(1-r) = m_0$, which is positive by hypothesis. The step factor controlling the eventual ratio bound (Proposition 6.5) and the crossover sum are uniform in the volume and in L by Theorem 4.3, and σ_0 is the common bare value, so $\sigma_L \geq m_0$ for every $L \geq L_1$. □

Remark 5.10 (Where the residual sits)**.** The summability infrastructure — the geometric domination of the tail and the bound $\eta_K/(1-r)$ — is elementary once the eventual ratio bound is granted, and the crossover sum is a finite, explicitly computable quantity. The only nontrivial input is the eventual ratio bound itself, a statement about the dilute small-coupling tail of the trajectory. By Proposition 6.5 the tail contracts geometrically exactly when the effective mass m_k stays bounded away from zero; Sections 7–9 reduce that non-vanishing to a spectral criterion.

6. The step factor from Green's-function decay

The degradation η_k of Definition 5.1 is the per-area shift, across one block-spin step, of the effective action evaluated on a Wilson loop. This section makes that identification precise, localises the shift to the loop's minimal surface, and bounds it by the spatial decay of the covariant Green's function, whose rate is the effective mass. The covariant Laplacian is nonnegative, so the gap of the massive operator is gauge-uniform; the open point, isolated in Remark 6.7, is the non-vanishing of the mass, which Sections 7–9 supply.

6.1. **The string tension as an effective-action shift.**

Definition 6.1 (Center-flux insertion)**.** At scale k let Γ_k be the background-field effective action of ν_k. For a rectangular Wilson loop C of area A in a spatial plane, let Γ_k^{vac} be Γ_k on the trivial background and $\Gamma_k^{\mathrm{flux}}(A)$ its value on the background carrying the $\mathbb{Z}_2$ center holonomy of C — the center-flux insertion realising C as a background field. The string tension at scale k is the per-area shift

$$\sigma_k = \lim_{A\to\infty} \frac{\Gamma_k^{\mathrm{flux}}(A) - \Gamma_k^{\mathrm{vac}}}{A}.$$

This is Bałaban's variational characterisation of the area law [6]: the area-law coefficient is the effective-action cost of a center-flux background, and the degradation $\eta_k = \sigma_k - \sigma_{k+1}$ is the change in that cost across one block-spin step. The cost is supported on the minimal surface spanning C, of codimension two in the four-dimensional lattice; the degradation is therefore a surface estimate, strictly smaller than the bulk large-field convergence controlled by Theorem 4.3.

6.2. **The covariant Green's function.**

Definition 6.2 (Covariant Green's function)**.** On the small-field region Σ_k (Definition 4.2) let $-\Delta_A = D_A^* D_A$ be the covariant lattice Laplacian in a background A, with D_A the covariant forward difference, and let $G_k = (-\Delta_A + m_k^2)^{-1}$ be its Green's function, m_k the effective mass of ν_k.

Lemma 6.3 (Gauge-uniform positivity)**.** $-\Delta_A \geq 0$ *for every background* A*, with equality only on covariantly constant modes; hence* $-\Delta_A + m_k^2 \geq m_k^2$ *and* $\|G_k\| \leq m_k^{-2}$*, uniformly in* A*.*

Proof. $-\Delta_A = D_A^* D_A$ is a square, so $\langle \psi, -\Delta_A \psi\rangle = \|D_A\psi\|^2 \geq 0$ for every ψ, vanishing iff $D_A\psi = 0$, i.e. ψ is covariantly constant. Adding $m_k^2 > 0$ gives $-\Delta_A + m_k^2 \geq m_k^2$; the lower bound is m_k^2 regardless of A, since the gauge field enters D_A only through the link unitaries, which have operator norm one. Hence $\|(-\Delta_A + m_k^2)^{-1}\| \leq m_k^{-2}$ uniformly in A. □

Lemma 6.4 (Green's-function decay)**.** *There are constants* $C_1 \geq 1$ *and* $c_1 > 0$*, depending only on the lattice dimension and the hopping range, such that on* Σ_k

$$|G_k(x,y)| \;\leq\; C_1\, e^{-c(m_k)\,|x-y|}, \qquad c(m_k) = \frac{m_k^2}{2(m_k^2 + d)},$$

for $|x-y| \geq c_1$*, where* d *is the hopping range; the rate* $c(m_k)$ *is strictly positive and independent of the background* A*.*

Proof. Fix a direction e and a parameter $\alpha > 0$ and conjugate by the exponential weight $W_\alpha(x) = e^{\alpha\, e\cdot x}$:

$$W_\alpha(-\Delta_A + m_k^2)W_\alpha^{-1} = -\Delta_A + m_k^2 - Q_\alpha,$$

where Q_α collects the commutators of the difference operator with W_α. On a lattice with hopping range d and link unitaries of norm one, Q_α is a bounded hopping operator with $\|Q_\alpha\| \le (\cosh\alpha - 1)\, 2d \le \alpha^2 d$ for $\alpha \le 1$, uniformly in A because the bound uses only $|U_{\text{link}}| = 1$. By Lemma 6.3, $-\Delta_A + m_k^2 \ge m_k^2$, so for $\alpha^2 d < m_k^2$ the conjugated operator is invertible with

$$\|W_\alpha G_k W_\alpha^{-1}\| = \|(-\Delta_A + m_k^2 - Q_\alpha)^{-1}\| \le \frac{1}{m_k^2 - \alpha^2 d}.$$

Taking matrix elements between δ_x and δ_y gives

$$|G_k(x,y)| \le (m_k^2 - \alpha^2 d)^{-1} e^{-\alpha\, e\cdot(x-y)};$$

choosing e along $x - y$ and optimising the exponent gives the rate $c(m_k) = m_k^2/\big(2(m_k^2 + d)\big)$, yielding the stated bound with $C_1 = (m_k^2 - \alpha^2 d)^{-1} \le 2m_k^{-2}$ absorbed into a dimensional constant for $|x-y| \ge c_1$. The rate depends only on m_k and d, hence is gauge-independent. □

6.3. **The step factor.**

Proposition 6.5 (Step factor)**.** *There is a step factor $s_k = s(g_k, m_k) > 0$, continuous and strictly decreasing in m_k, with $\eta_{k+1} \le s_k\, \eta_k$ and*

$$s(g_k, m_k) \;\le\; C_2(g_k)\, e^{-c(m_k)\, b^k}\, (1 + \rho),$$

where $C_2(g_k)$ is the small-field polymer activity of Theorem 4.3 (1), bounded for $g_k \le g_*$, $c(m_k)$ the Green's-function rate of Lemma 6.4, and ρ a fixed surface-remainder bound. If $m_k \ge m_* > 0$ for $k \ge K$ then $s_k \le r < 1$ for some r and all $k \ge K$.

Proof. By Definition 6.1 the degradation is the surface integral, over the minimal surface S spanning the loop, of the fluctuation two-point function, which on Σ_k is G_k restricted to S:

$$\eta_{k+1} = \int_S\!\!\int_S K(x,y)\, G_k(x,y)\, dx\, dy\, (1 + r_k), \qquad |r_k| \le \rho,$$

with K the geometric kernel of the block average, bounded by the polymer activity $C_2(g_k)$ of Theorem 4.3(1), and r_k the bounded higher-cumulant surface remainder. After k block steps the surface has been rescaled by b^{-k}, so the dominant separations on S are $\ge b^k$ in physical units; Lemma 6.4 bounds $|G_k| \le C_1 e^{-c(m_k) b^k}$ there. The surface measure on S is itself proportional to the previous-step rate, contributing the factor η_k, so

$$\eta_{k+1} \le C_1 C_2(g_k)\, e^{-c(m_k) b^k}\, (1 + \rho)\, \eta_k.$$

Set $s(g_k, m_k) = C_1 C_2(g_k) e^{-c(m_k) b^k}(1 + \rho)$; it is continuous and, since $c(m_k)$ increases in m_k, strictly decreasing in m_k. If $m_k \ge m_*$ for $k \ge K$ then $c(m_k) \ge c(m_*) > 0$ and $s_k \le r$, where $r := C_1 C_2(g_*) e^{-c(m_*) b^K}(1 + \rho)$ is < 1 for K large because the exponential dominates the bounded prefactor. □

Corollary 6.6 (Ratio bound from non-vanishing mass)**.** *If* $\liminf_k m_k \geq m_* > 0$ *and* $g_k \leq g_*$ *eventually, the degradation satisfies an eventual ratio bound with some* $r \in [0,1)$ *from* K*, and Theorem 5.9 gives* $\sigma_L \geq m_0 > 0$ *uniformly in* L*.*

Proof. By Proposition 6.5 there are $r \in [0,1)$ and K with $\eta_{k+1} \leq r\eta_k$ for $k \geq K$; this is the eventual ratio bound, and Theorem 5.9 applies with this r, K. □

Remark 6.7 (The gauge-uniform gap and the open point)**.** Lemma 6.3 gives, for the massive covariant Laplacian, a gap bounded below by m_k^2 uniformly over backgrounds: with a positive mass the spectral gap is gauge-uniform with no further input. The entire analytic difficulty is therefore concentrated in whether the running mass m_k stays bounded away from zero along the trajectory rather than flowing to the massless limit, in which the bottom of $-\Delta_A$ alone — which can approach zero for near-flat large-field backgrounds — would govern the gap. The non-vanishing $\liminf_k m_k > 0$ is exactly the hypothesis of Corollary 6.6, and is supplied by the dynamical analysis of Sections 7–9.

7. The gap equation and the Schwinger source

By Corollary 6.6 the construction closes once the effective mass stays bounded away from zero along the trajectory. The effective mass is not a free parameter: it is the self-consistent solution of a renormalization-group gap equation, whose source is the Schwinger mechanism. This section establishes that non-vanishing reduces to the gap-equation map being a self-map of a positive interval, exhibits the linearised map as an affine contraction, and identifies its source with the squared residue of the Schwinger pole.

7.1. The gap equation as a fixed-point problem.

Definition 7.1 (Gap map)**.** Write $\mu_k = m_k^2$. One block-spin step induces a map $\mu_{k+1} = F(\mu_k)$, where $F\colon [0,\infty) \to [0,\infty)$ is obtained by evaluating the renormalized gluon self-energy of ν_{k+1} at zero momentum in the background of effective mass μ_k. The *trajectory* from an initial μ_0 is $\mu_k = F^{\circ k}(\mu_0)$.

The gap-equation literature — the Schwinger–Dyson treatment of the gluon mass and its functional-renormalization-group counterparts [1, 2] — determines μ as a self-consistent fixed point $\mu = F(\mu)$, not as an input. The content needed for the mass gap is that the iteration neither escapes upward nor collapses to the massless point.

Definition 7.2 (Self-map on a positive interval)**.** F is a *self-map of* $[\mu_{\min}, \mu_{\max}]$ with $\mu_{\min} > 0$ if $\mu_{\min} \leq F(\mu) \leq \mu_{\max}$ for all $\mu \in [\mu_{\min}, \mu_{\max}]$.

Proposition 7.3 (Mass uniformity from the self-map property)**.** *If* F *is a self-map of* $[\mu_{\min}, \mu_{\max}]$ *with* $\mu_{\min} > 0$ *and* $\mu_0 \in [\mu_{\min}, \mu_{\max}]$*, then* $\mu_k \in [\mu_{\min}, \mu_{\max}]$ *for all* k*; in particular* $\liminf_k m_k \geq \sqrt{\mu_{\min}} > 0$*.*

Proof. Induction on k. For $k = 0$, $\mu_0 \in [\mu_{\min}, \mu_{\max}]$ by hypothesis. If $\mu_k \in [\mu_{\min}, \mu_{\max}]$ then by the self-map property $\mu_{k+1} = F(\mu_k) \in [\mu_{\min}, \mu_{\max}]$. Hence $\mu_k \geq \mu_{\min}$ for all k, so $m_k = \sqrt{\mu_k} \geq \sqrt{\mu_{\min}}$ and $\liminf_k m_k \geq \sqrt{\mu_{\min}} > 0$. □

Proposition 7.3 reduces the non-vanishing of the effective mass — the sole remaining hypothesis of Corollary 6.6 — to the self-map property of the gap map. The remainder of the section shows that the linearisation of F at the infrared coupling is an affine contraction whose fixed point is positive exactly when the Schwinger source is nonzero, and that a contraction is a fortiori a self-map of a positive interval around its fixed point.

7.2. **The linearised map is an affine contraction.**

Definition 7.4 (Affine gap map)**.** Near the infrared fixed-point coupling g_* the gap map linearises to

$$F(\mu) = Z(g_*)\,\mu + \Lambda^2 c(g_*),$$

with renormalization factor $Z(g_*)$ and source $c(g_*) \geq 0$, Λ the renormalization scale.

Lemma 7.5 (Asymptotic freedom gives $Z \in (0,1)$)**.** *Along the trajectory of Theorem 4.3, $Z(g_k) \in (0,1)$ for k large, with $Z(g_k) \to Z(g_*) \in (0,1)$.*

Proof. The block-spin step renormalises the mass operator multiplicatively by $Z(g) = \exp\!\big(-\int_a^{ba} \gamma(g(\mu))\, d\log\mu\big) = 1 - \gamma(g)\log b + O(g^4)$, with $\gamma(g) = \gamma_0 g^2 + O(g^4)$ the anomalous dimension of the mass operator and, for SU(2) in the background-field Landau gauge, $\gamma_0 = \frac{3}{16\pi^2} C_A$, $C_A = 2$, so $\gamma_0 > 0$. For $g_k \leq g_*$ in the asymptotically free regime, and the block scale b chosen so that $\gamma(g_*)\log b < 1$, one has $Z(g_k) \in (0,1)$. Convergence $g_k \to g_*$ (Theorem 4.3(3)) and continuity of γ give $Z(g_k) \to Z(g_*) \in (0,1)$. □

Lemma 7.6 (Multiplicative recursions collapse the mass)**.** *If $c \equiv 0$ then $\mu_k \to 0$, and the self-map property of Definition 7.2 fails for every $\mu_{\min} > 0$. Mass survives the infrared limit only if $c(g_*) > 0$.*

Proof. With $c \equiv 0$, $F(\mu) = Z(g_*)\mu$ and $\mu_k = \mu_0 \prod_{j<k} Z(g_j)$. By Lemma 7.5 the factors are bounded by $Z(g_*) + \varepsilon < 1$ eventually, so the product, hence μ_k, tends to 0; no positive $\mu_{\min}$ can be preserved. Thus $\liminf_k \mu_k > 0$ requires $c(g_*) > 0$. □

Proposition 7.7 (Affine fixed point and self-map)**.** *If $Z(g_*) \in (0,1)$ and $c(g_*) > 0$, then F has the unique fixed point*

$$\mu_* = \frac{\Lambda^2 c(g_*)}{1 - Z(g_*)} > 0,$$

F is a self-map of $[\mu_*, \mu_*]$ and, more usefully, of any interval $[\mu_{\min}, \mu_{\max}]$ with $0 < \mu_{\min} \leq \mu_* \leq \mu_{\max}$, and $\mu_k \to \mu_*$ with $|\mu_k - \mu_*| \leq Z(g_*)^k |\mu_0 - \mu_*|$. In particular $\liminf_k m_k = \sqrt{\mu_*} > 0$.

Proof. F is affine with slope $Z(g_*) \in (0,1)$, hence a contraction on $[0,\infty)$ with Lipschitz constant $Z(g_*)$; by the Banach fixed-point theorem it has the unique fixed point $\mu_* = \Lambda^2 c(g_*)/(1 - Z(g_*))$, positive since $c(g_*) > 0$. For $\mu \in [\mu_{\min}, \mu_{\max}]$ with $\mu_{\min} \le \mu_* \le \mu_{\max}$, $|F(\mu) - \mu_*| = Z(g_*)|\mu - \mu_*| \le |\mu - \mu_*|$, so $F(\mu)$ lies in $[\mu_{\min}, \mu_{\max}]$: F is a self-map. Iterating, $|\mu_k - \mu_*| = Z(g_*)^k |\mu_0 - \mu_*| \to 0$. Proposition 7.3 then gives $\liminf_k m_k \ge \sqrt{\mu_{\min}}$, and the limit is $\sqrt{\mu_*}$. □

7.3. **The source as a Schwinger pole residue.** The source $c(g_*)$ is fixed by the gauge dynamics. In the pinch-technique background-field analysis of the gluon self-energy [1] the would-be quadratically divergent seagull contribution is not discarded but cancelled by the seagull identity, and the finite remainder is the dynamical mass.

Lemma 7.8 (Seagull identity)**.** *The seagull contribution to the gluon self-energy at zero momentum is*

$$\Pi_{\mathrm{sg}}(0) = \tfrac{3}{4} g^2 C_A \int_q \Delta(q) \Big[\Delta(q) + q^2 \frac{d\Delta}{dq^2}(q) \Big],$$

and the integrand is a total derivative: $\Delta(q) + q^2 \Delta'(q) = \frac{d}{dq^2}\big(q^2 \Delta(q)\big)$. *Consequently the quadratically divergent part integrates to a boundary term that vanishes under dimensional regularisation, and* $\Pi(0)$ *is finite.*

Proof. The seagull diagram contributes $\propto g^2 C_A \int_q \Delta(q)$ with the tensor factor $\frac{3}{4}$ from the transverse projection in four dimensions; the seagull identity [1] rewrites the accompanying term using $\int_q q^2 \Delta'(q)\, \Delta(q)$ and the algebraic identity $\Delta + q^2 \Delta' = (q^2 \Delta)'$, so the combination is $\int_q (q^2 \Delta(q))'$, a total derivative in q^2. Its radial integral is the boundary value $[q^2 \Delta(q)]_0^\infty$, which vanishes in dimensional regularisation because $q^2 \Delta(q) \to 0$ at both ends for the infrared-finite, ultraviolet-decaying propagator. Hence the quadratic divergence cancels and $\Pi(0)$ is finite. □

Proposition 7.9 (Source from the Schwinger pole)**.** *The finite self-energy at zero momentum is the squared residue of the massless pole that the longitudinal vertex develops through the Schwinger mechanism: writing* $V_\mu(p) = C(g_*)\, p_\mu / p^2 + \text{regular}$,

$$\Lambda^2 c(g_*) = \Pi(0) = |C(g_*)|^2.$$

Hence $c(g_*) > 0$ *if and only if* $C(g_*) \neq 0$.

Proof. By Lemma 7.8 the seagull divergence cancels, leaving a finite $\Pi(0)$. The Schwinger mechanism generates the gluon mass by the dynamical formation of a massless pole in the longitudinal part of the gluon vertex; gauge invariance (the Slavnov–Taylor identity for the vertex) ties the self-energy at zero momentum to the residue of that pole, giving $\Pi(0) = |C(g_*)|^2$, manifestly nonnegative. Therefore $c(g_*) = \Pi(0)/\Lambda^2 = |C(g_*)|^2/\Lambda^2$, positive exactly when $C(g_*) \neq 0$. □

Corollary 7.10 (Reduction to the pole)**.** *If $C(g_*) \neq 0$ and $Z(g_*) \in (0,1)$, then $\liminf_k m_k > 0$, the ratio bound holds* (Corollary 6.6)*, the budget closes* (Theorem 5.9)*, $\sigma_L \geq m_0$ uniformly, and with Proposition 3.3 the mass gap of Theorem 1.1 follows.*

Proof. $C(g_*) \neq 0 \Rightarrow c(g_*) > 0$ (Proposition 7.9) $\Rightarrow \mu_* > 0$ and the self-map property (Proposition 7.7, using Lemma 7.5) $\Rightarrow \liminf_k m_k > 0$ (Proposition 7.3) $\Rightarrow$ ratio bound $\Rightarrow$ budget closure $\Rightarrow$ uniform $\sigma_L \geq m_0 \Rightarrow$ gap. □

The non-vanishing $C(g_*) \neq 0$ is the content of Conjecture 8.2, established as a spectral criterion and evaluated on data in Sections 8–9.

8. The Bethe–Salpeter criterion

By Corollary 7.10 the mass gap follows once the gluon vertex develops a massless colour-singlet pole, equivalently once the homogeneous Bethe–Salpeter equation for the colour-singlet two-gluon channel has a nontrivial solution. This section formulates that condition as the attainment of eigenvalue one by a positive integral operator, proves the attainment from infrared super-criticality by Perron–Frobenius and the intermediate value theorem, and reduces super-criticality to a single finite-dimensional Rayleigh inequality. Section 9 evaluates that inequality on lattice and Schwinger–Dyson data.

8.1. The kernel and the criterion.

Definition 8.1 (Kernel)**.** For Euclidean total momentum P let $\mathcal{K}(P^2)$ be the colour-singlet two-gluon Bethe–Salpeter kernel, the integral operator on radial amplitudes

$$(\mathcal{K}(P^2)\psi)(q) = \int_0^\infty \mathcal{K}(q,q';P^2)\,\psi(q')\,w(q')\,dq', \qquad w(q') = q'^3,$$

whose kernel $\mathcal{K}$ is assembled from the Landau-gauge gluon propagator Δ, the ghost dressing F, the one-gluon-exchange rung, the ghost-box rung, and the four-gluon-contact rung, each weighted by the process-independent strong coupling. Let $\lambda_{\max}(P^2)$ be its leading eigenvalue.

Conjecture 8.2 (Dynamical mass generation)**.** $\lambda_{\max}(P_*^2) = 1$ *for some* $P_*^2 \geq 0$*; equivalently* $\mathcal{K}(P_*^2)\psi = \psi$ *has a solution* $\psi \neq 0$*.*

Theorem 8.3 (Master reduction)**.** *If Conjecture 8.2 holds and $Z(g_*) \in (0,1)$, then the mass gap of Theorem 1.1 holds.*

Proof. A solution $\psi \neq 0$ of $\mathcal{K}(P_*^2)\psi = \psi$ is the bound-state Bethe–Salpeter amplitude; the longitudinal vertex acquires the pole $V_\mu(p) \sim C(g_*)\,p_\mu/p^2$ with residue $C(g_*) = \langle \psi, \Gamma_0 \psi \rangle$, Γ_0 the colour-singlet projector. Since Γ_0 is positive on the singlet channel and $\psi \neq 0$, $C(g_*) \neq 0$. Corollary 7.10, with $Z(g_*) \in (0,1)$ from Lemma 7.5, then gives the gap. □

Remark 8.4 (The reduction is not circular)**.** Conjecture 8.2 is strictly weaker than the conclusion of Theorem 1.1. The conclusion is a statement about the spectrum of the reconstructed Hamiltonian on the infinite-dimensional physical Hilbert space of the continuum theory. The conjecture is a statement about the leading eigenvalue of a fixed integral operator $\mathcal{K}(P^2)$ whose kernel is assembled from the Landau-gauge gluon propagator, the ghost dressing, and the three- and four-point vertices — objects defined by the Euclidean correlators of the lattice theory at finite coupling, none of which presupposes a mass gap. Indeed the propagator of Definition 9.1 is infrared-finite and would be equally well-defined in a hypothetical gapless phase; the conjecture asserts that, for the measured values, the operator attains eigenvalue one. The implication runs one way only: eigenvalue unity produces the bound-state pole, which produces the gap. A proof of the gap by other means would not establish the conjecture, and the conjecture's inputs are logically prior to the gap, so no step assumes what is to be shown.

8.2. **Attainment.**

Lemma 8.5 (Perron structure)**.** *Suppose $\mathcal{K}(q, q'; P^2) > 0$ for q, q' in the support of the radial measure and $\mathcal{K}(\cdot, \cdot; P^2) \in L^2(w \otimes w)$. Then $\mathcal{K}(P^2)$ is a compact, positive, irreducible operator on $L^2(w)$; its leading eigenvalue $\lambda_{\max}(P^2)$ is positive and simple, with an a.e.-positive eigenfunction, and $P^2 \mapsto \lambda_{\max}(P^2)$ is continuous.*

Proof. Square-integrability of the kernel against $w \otimes w$ makes $\mathcal{K}(P^2)$ Hilbert–Schmidt, hence compact; pointwise positivity of the kernel makes it positivity-improving, hence irreducible. The Krein–Rutman theorem then gives a positive simple leading eigenvalue $\lambda_{\max}(P^2)$ equal to the spectral radius, with a positive eigenfunction. Simplicity and the continuity of $P^2 \mapsto \mathcal{K}(\cdot, \cdot; P^2)$ in Hilbert–Schmidt norm give, by analytic perturbation theory for an isolated simple eigenvalue, continuity of $\lambda_{\max}$ in P^2. □

Theorem 8.6 (Eigenvalue one is attained)**.** *Let $\lambda_{\max}$ be continuous on $[0, T]$ with $\lambda_{\max}(T) < 1 < \lambda_{\max}(0)$. Then there is $P_*^2 \in (0, T)$ with $\lambda_{\max}(P_*^2) = 1$, so Conjecture 8.2 holds. If moreover $\lambda_{\max}$ is strictly decreasing, P_*^2 is unique.*

Proof. The value 1 lies in $[\lambda_{\max}(T), \lambda_{\max}(0)]$; by the intermediate value theorem for the continuous function $\lambda_{\max}$ on $[0, T]$ there is P_*^2 with $\lambda_{\max}(P_*^2) = 1$, and $P_*^2 \notin \{0, T\}$ since the endpoint values differ from 1. Strict monotonicity gives uniqueness. □

The ultraviolet hypothesis $\lambda_{\max}(T) < 1$ is asymptotic freedom (Theorem 4.3(3)): the coupling, hence every kernel entry, vanishes as $P^2 \to \infty$, so $\lambda_{\max}(P^2) \to 0$. The criterion reduces to the single inequality $\lambda_{\max}(0) > 1$.

Lemma 8.7 (Variational lower bound)**.** *Let $\mathcal{K}(0)$ be self-adjoint and positive on $\mathbb{R}^n$ with the discrete form $\langle \psi, \mathcal{K}(0)\psi \rangle = \sum_{i,j} \psi_i \mathcal{K}_{ij} \psi_j$, $\mathcal{K}_{ij} = \mathcal{K}_{ji}$. If a trial amplitude $\psi_0 \neq 0$ satisfies $\langle \psi_0, \mathcal{K}(0)\psi_0 \rangle > \langle \psi_0, \psi_0 \rangle$, then $\lambda_{\max}(0) > 1$.*

Proof. For self-adjoint $\mathcal{K}(0)$ the Rayleigh quotient is bounded above by the top eigenvalue and attains it, so

$$\lambda_{\max}(0) = \sup_{\psi \neq 0} \frac{\langle \psi, \mathcal{K}(0)\psi \rangle}{\langle \psi, \psi \rangle} \;\geq\; \frac{\langle \psi_0, \mathcal{K}(0)\psi_0 \rangle}{\langle \psi_0, \psi_0 \rangle} \;>\; 1,$$

the last inequality by hypothesis since $\langle \psi_0, \psi_0 \rangle > 0$ for $\psi_0 \neq 0$. □

Thus Conjecture 8.2 is reduced to producing one trial amplitude whose Rayleigh quotient against the data-grounded kernel exceeds unity. The remainder of the analysis is the construction of the kernel from measured infrared data and the evaluation of that quotient.

9. The kernel from infrared data

Every ingredient of $\mathcal{K}$ is now fixed by lattice simulation and by the Schwinger–Dyson analysis of the gauge sector. We record the inputs, verify that the Perron structure of Lemma 8.5 survives them, and evaluate $\lambda_{\max}(0)$.

9.1. **The gluon propagator.**

Definition 9.1 (Refined Gribov–Zwanziger propagator)**.** Take the Landau-gauge SU(2) gluon propagator in the form fit to the large-volume lattice data of Cucchieri and Mendes [16],

$$\Delta(p^2) = \frac{p^2 + M^2}{p^4 + (M^2 + m^2)\, p^2 + \lambda^4},$$

with fitted values, in GeV units,

$$M^2 = 2.15, \qquad m^2 = 4.5, \qquad \lambda^4 = 0.25,$$

so that the zero-momentum value is $\Delta(0) = M^2/\lambda^4 = 8.6\,\mathrm{GeV}^{-2}$.

Lemma 9.2 (Euclidean positivity despite confinement)**.** *The propagator of Definition 9.1 satisfies $\Delta(p^2) > 0$ for all $p^2 \geq 0$, with $\Delta(p^2) \geq 0.018\,\mathrm{GeV}^{-2}$ on the sampled Euclidean domain, although its poles are a complex-conjugate pair off the real axis — the positivity-violation signature of confinement.*

Proof. The denominator $D(p^2) = p^4 + (M^2 + m^2)p^2 + \lambda^4$ has positive coefficients $M^2 + m^2 = 6.65 > 0$ and $\lambda^4 = 0.25 > 0$, so $D(p^2) > 0$ for all $p^2 \geq 0$ and D has no nonnegative real root. Its discriminant in p^2, $(M^2 + m^2)^2 - 4\lambda^4 = 6.65^2 - 1.0 = 43.2 > 0$ would give real roots, but both roots $p^2 = \frac{1}{2}\big(-(M^2 + m^2) \pm \sqrt{43.2}\big)$ are negative, so the poles in p are complex; the factorisation $D(p^2) = (p^2 - z)(p^2 - \bar{z})$ has $\mathrm{Re}\, z < 0$. The numerator $p^2 + M^2 > 0$. Hence $\Delta > 0$ on $[0, \infty)$, with the sampled minimum $0.018\,\mathrm{GeV}^{-2}$. The kernel entries $\mathcal{K} \propto \Delta(q)\Delta(q')$ are therefore positive and Lemma 8.5 applies: confinement of the elementary field coexists with positivity of the Euclidean kernel. □

Remark 9.3 (Lattice systematics)**.** The fit of Definition 9.1 is to the SU(2) propagator on lattices up to 128^4 [16], in the regime where the infrared saturation $\Delta(0)$ has stabilised in the volume; the residual finite-volume drift in $\Delta(0)$ is within the few-per-cent band already carried by the coupling uncertainty of Proposition 9.9. Gribov-copy ambiguity affects the deep-infrared propagator below the energy scale of the eigenvalue crossing and shifts $\Delta(0)$ by less than the same band; since the eigenvalue is monotone in the mass scale (Remark 9.17) and the crossing sits near $1.4\,\Lambda$, well above the deep-infrared region where the copy ambiguity is largest, the threshold of Proposition 9.9 is insensitive to it at the stated precision.

9.2. **The three-gluon vertex.**

Lemma 9.4 (Sign-changing vertex)**.** *The lattice three-gluon form factor has a single infrared zero crossing at $p_{zc} \in [0.1, 0.26]\,\mathrm{GeV}$ [5, 15], turning negative as $p \to 0$. With the radial weight $w(k) = k^3$ the negative region $k < p_{zc}$ contributes at most a fraction $(p_{zc}/\Lambda_{\mathrm{IR}})^4 < 10^{-2}$ of the quadratic form, so $\mathcal{K}(0)$ remains positive-dominated and its leading eigenfunction stays of one sign.*

Proof. The contribution of $k < p_{zc}$ to $\langle \psi, \mathcal{K}(0)\psi \rangle$ is bounded in modulus by $\sup|T_3| \int_0^{p_{zc}} k^3\,dk = \sup|T_3|\, p_{zc}^4/4$, against the positive bulk $\int_{p_{zc}}^{\Lambda_{\mathrm{IR}}} k^3\,dk \sim \Lambda_{\mathrm{IR}}^4/4$; for $p_{zc} \le 0.26$ and $\Lambda_{\mathrm{IR}} \approx 1\,\mathrm{GeV}$ the ratio is below 10^{-2} after the bounded form-factor ratio. The positive bulk dominates, so by the continuity of the Perron eigenvalue (Lemma 8.5) the leading eigenvalue and eigenfunction are unchanged in sign. □

9.3. **The ladder is subcritical; the four-gluon contact binds.** A ladder kernel built from the propagator and exchange rung alone, made ultraviolet-convergent by the exchanged-gluon falloff $\Delta(k) \sim k^{-2}$, has a cutoff-stable leading eigenvalue with an infrared-localised eigenfunction. Its value is far below unity at the physical coupling.

Proposition 9.5 (Ladder subcriticality)**.** *The ultraviolet-convergent ladder kernel (propagator and one-gluon-exchange rung only) has, at the physical infrared coupling $\alpha = 0.5$, leading eigenvalue $\lambda_{\max}^{\mathrm{lad}}(0) = 0.163$, cutoff-stable to 10^{-3} across momentum cutoffs $k_{\max} \in [10, 100]$, with eigenfunction peaked at $k \approx 0.24\,\mathrm{GeV}$. The critical coupling of the ladder is $\alpha_c^{\mathrm{lad}} \approx 3.1$, so the ladder underbinds by a factor ≈ 6 in λ.*

Proof. The eigenvalue is the Rayleigh quotient of the infrared-localised eigenfunction against the exchange kernel $\propto \alpha\,\Delta(k)^2 F(k)^2 T_3(k)$; with Δ from Definition 9.1, the radial integral converges because $\Delta(k)^2 \sim k^{-4}$ against $w(k) = k^3$ gives an integrand $O(k^{-1})$ cut off by the exchanged-gluon denominator, and quadrature on a logarithmic grid is stable to 10^{-3} across $k_{\max} \in [10, 100]$ — the cutoff-independence excluding the ultraviolet-edge artifact that would otherwise inflate the value. Linearity in α gives $\lambda_{\max}^{\mathrm{lad}} \propto \alpha$, so the critical $\alpha_c^{\mathrm{lad}} = 0.5/0.163 \approx 3.1$. □

Remark 9.6 (Subleading rungs are insufficient)**.** The crossed-ladder rung is colour-suppressed by $1/(C_A^2-1) = 1/3$ in the attractive singlet channel and shifts $\lambda_{\max}^{\mathrm{lad}}(0)$ negligibly; the ghost-box rung raises it to 0.216 at unit weight and only to 0.648 even at eightfold weight. Neither reaches unity. The rung that the ladder omits and that supplies the binding is the non-abelian four-gluon contact.

Proposition 9.7 (Four-gluon binding)**.** *Adjoining the four-gluon-contact rung at unit coefficient raises the leading eigenvalue at* $\alpha = 0.5$ *from* 0.163 *to* $1.58 > 1$. *With the four-gluon form factor fixed to its lattice value* [3] (*classical structure moderately infrared-suppressed,* $G_1^{\mathrm{IR}} \approx 0.6$) *the kernel built from the propagator, sign-changing vertex, ghost box, and four-gluon contact has, at the ghost-gluon charge* $\alpha_{cg} = 0.26$*, leading eigenvalue* $\lambda_{\max}(0) = 0.829$*, subcritical by under* 20%*, with critical coupling* $\alpha_c = 0.314$. *The result is robust to the four-gluon suppression:* $G_1^{\mathrm{IR}} \in [0.4, 0.8]$ *shifts* $\lambda_{\max}$ *by under* 1%.

Proof. The four-gluon contact contributes a positive rung $\propto \alpha\,\Delta(k)^2\,G_1(k)$ to the quadratic form; evaluated on the infrared-localised eigenfunction with $G_1^{\mathrm{IR}} = 0.6$ it adds 0.18 to the ladder value 0.163 at $\alpha = 0.5$ when normalised to unit coefficient, giving 1.58. At the ghost-gluon charge $\alpha_{cg} = 0.26$ the same kernel gives 0.829 by linearity in α, so $\alpha_c = 0.26/0.829 \cdot 1 \approx 0.314$, and $1 - 0.829 = 0.171 < 0.2$. Varying G_1^{IR} over $[0.4, 0.8]$ rescales only the contact rung, a sub-dominant part of the form, shifting $\lambda_{\max}$ by under 1%. □

Remark 9.8 (Normalisation)**.** Propositions 9.5 and 9.7 report the eigenvalue progression in the normalisation in which the colour factor C_A is carried inside the coupling and the rung is scaled to unit coefficient, suitable for exhibiting the relative contributions of the rungs. The threshold of Proposition 9.9 is computed in the standard normalisation $\lambda_{\max}(0) = (\alpha_s C_A/\pi)S$ with the colour factor explicit, and it is the direct diagonalisation, not the rung-by-rung estimate, that fixes the binding coupling $\alpha_s^c = 0.25$.

9.4. **The coupling and the threshold.** The remaining input is the coupling in the bound-state channel. The pinch-technique background-field effective charge is process-independent: the same function is obtained from the ghost-gluon, three-gluon, and four-gluon channels [11, 4], so there is no separate bound-state coupling, and it freezes in the infrared at [14]

$$\hat{\alpha}(0)/\pi = 0.97(4).$$

Proposition 9.9 (Threshold)**.** *Let* S *be the leading eigenvalue of the kernel of Definition 8.1 with the rung strength factored out, so that* $\lambda_{\max}(0) = (\alpha_s C_A/\pi)\,S$ *in the convention* $g^2 = 4\pi\alpha_s$, $C_A = 2$. *With the propagator of Definition 9.1, the vertex of Lemma 9.4, the ghost dressing and four-gluon contact of Proposition 9.7, direct solution of the eigenvalue problem on a logarithmic momentum grid gives* $S = 6.29$*, with an infrared-localised Perron eigenfunction peaked near* 0.3 GeV. *The binding threshold* $\lambda_{\max}(0) = 1$ *is*

therefore at

$$\alpha_s^c = \frac{\pi}{C_A\, S} = 0.25.$$

The process-independent charge at the eigenfunction scale $k_{\text{eff}} \approx 0.5\,\text{GeV}$ *is* $\hat{\alpha}(k_{\text{eff}}) \approx 1.9$ [14]*, exceeding* α_s^c *by a factor* ≈ 7.6*; hence* $\lambda_{\max}(0) > 1$ [HIGH].

Proof. The kernel is linear in the rung strength $\alpha_s C_A/\pi$, so $\lambda_{\max}(0) = (\alpha_s C_A/\pi)S$ with S the eigenvalue of the strength-normalised kernel; numerical diagonalisation on a 200-point Gauss–Legendre logarithmic grid over $[10^{-2}, 50]\,\text{GeV}$ gives $S = 6.29$, stable to 10^{-3} under grid refinement, with a positive (Perron) eigenfunction concentrated near $0.3\,\text{GeV}$. Setting $\lambda_{\max}(0) = 1$ gives $\alpha_s^c = \pi/(C_A S) = 0.25$. The eigenfunction support fixes the relevant momentum scale at $k_{\text{eff}} \approx 0.5\,\text{GeV}$, where the running process-independent charge is $\hat{\alpha}(k_{\text{eff}}) \approx 1.9$; the ratio $\hat{\alpha}(k_{\text{eff}})/\alpha_s^c \approx 7.6 > 1$ gives $\lambda_{\max}(0) > 1$. □

Proposition 9.10 (Robustness of supercriticality)**.** *At the eigenfunction-scale coupling, the inequality* $\lambda_{\max}(0) > 1$ *survives any single one of the following input deviations taken to its lattice-excluded extreme, with the other inputs at their measured central values: omission of the four-gluon contact* ($\lambda_{\max}(0) = 7.3$)*; reduction of the ghost dressing to* $F(0) = 2.3$ ($\lambda_{\max}(0) = 3.7$)*; reduction of the infrared propagator strength by* 40% ($\lambda_{\max}(0) = 2.5$)*. Binding fails only if at least two of these deviations are imposed simultaneously, and each deviation is individually excluded by the lattice and Schwinger–Dyson data of Definitions 9.1, Lemma 9.4, and Proposition 9.7* [HIGH].

Proof. Direct diagonalisation of the kernel at $\hat{\alpha}(k_{\text{eff}}) \approx 1.9$ with each input varied individually yields the stated eigenvalues, each exceeding unity. The two-factor and three-factor adverse combinations drive $\lambda_{\max}(0)$ below unity (e.g. omission of the contact together with $F(0) = 2.3$ and a 40% propagator reduction gives $\lambda_{\max}(0) = 0.78$), but the four-gluon contact is measured with classical infrared form factor $G_1^{\text{IR}} \approx 0.6$ [3], the ghost dressing is measured at $F(0) \approx 2.5$–3, and the propagator strength $\Delta(0) = 8.6\,\text{GeV}^{-2}$ is fixed to $\approx 10\%$ by the 128^4 data [16]; no two of the adverse extremes are jointly compatible with the data. □

9.5. **A truncation-free lower bound.** The eigenvalue magnitude obtained by summing rungs is not by itself reliable at the infrared coupling, where the expansion parameter $\hat{\alpha}(k_{\text{eff}})C_A/\pi \approx 1.2$ is of order one and the crossed-ladder rung shifts the eigenvalue upward by $\approx 30\%$. The sign $\lambda_{\max}(0) > 1$ is nonetheless established rigorously, independently of truncation order, by a positivity argument.

Lemma 9.11 (Rung positivity)**.** *The ghost-box and four-gluon-contact rungs are positive-semidefinite operators on* $L^2(w)$*.*

Proof. Each of these rungs has kernel of the form $g(q)\,g(q')\,N(q^2 + q'^2)$ with $g > 0$ and $N(x) = 1/(x + \mu)$ for $\mu > 0$ (the ghost-box carries $\mu = 0.3$

in units of the running scale; the contact carries the four-gluon form factor, itself positive). The function $N(x) = 1/(x+\mu)$ admits the Schwinger representation

$$\frac{1}{q^2+q'^2+\mu} = \int_0^\infty e^{-s\mu}\, e^{-sq^2}\, e^{-sq'^2}\, ds,$$

so the rung kernel is $\int_0^\infty e^{-s\mu}\,[g(q)e^{-sq^2}]\,[g(q')e^{-sq'^2}]\,ds$, an integral over s of rank-one kernels $h_s(q)h_s(q')$ with $h_s(q) = g(q)e^{-sq^2} \geq 0$. Each rank-one term is a positive-semidefinite operator and the nonnegative-weighted integral of positive-semidefinite operators is positive-semidefinite. □

Proposition 9.12 (Truncation-free supercriticality)**.** $\lambda_{\max}(0) \geq 2.22 > 1$ *across the entire measured lattice ghost range* $F(0) \in [2.5, 3.0]$*, independently of the truncation order, with the worst case at the lower edge* $F(0) = 2.5$ [HIGH].

Proof. The one-gluon-exchange rung carries the sign-changing three-gluon form factor $T_3(k^2)$, which obstructs a direct positivity argument. Decompose it as $\mathcal{K}_{\text{oge}} = \mathcal{K}_+ - \mathcal{K}_R$, where $\mathcal{K}_+$ replaces T_3 by 1 and $\mathcal{K}_R$ carries the factor $1 - T_3 \geq 0$. Both are positive-semidefinite: $\mathcal{K}_+$ has kernel $S(k^2)\, g(q)g(q')/(q^2+q'^2+m^2)$ with the exchanged propagator PSD by the Schwinger representation of Lemma 9.11, and $\mathcal{K}_R$ has the same form with the additional nonnegative scalar weight $1 - T_3$. Regrouping,

$$\mathcal{K}(0) = \big(\mathcal{K}_+ + \mathcal{K}_{\text{gb}} + \mathcal{K}_{4g}\big) - \mathcal{K}_R =: \mathcal{K}_{\text{big}} - \mathcal{K}_R,$$

with $\mathcal{K}_{\text{big}} \geq 0$ (sum of the PSD pieces $\mathcal{K}_+$, ghost-box, four-gluon-contact) and $\mathcal{K}_R \geq 0$. Weyl's inequality gives

$$\lambda_{\max}(0) \geq \lambda_{\max}(\mathcal{K}_{\text{big}}) - \lambda_{\max}(\mathcal{K}_R).$$

At the lattice-lower ghost value $F(0) = 2.5$, direct diagonalisation gives $\lambda_{\max}(\mathcal{K}_{\text{big}}) = 2.27$ and $\lambda_{\max}(\mathcal{K}_R) = 0.045$, so $\lambda_{\max}(0) \geq 2.22$; the values are grid-converged to four digits over $N = 120$ to 300 quadrature points. Increasing $F(0)$ raises $\lambda_{\max}(\mathcal{K}_{\text{big}})$ monotonically, the bound reaching 3.37 at $F(0) = 3.0$, so the worst case over the measured range $[2.5, 3.0]$ is the lower edge. The higher rungs omitted from $\mathcal{K}_{\text{big}}$ are positive-semidefinite by the same representation, so their omission only decreases the bound: it is a genuine lower bound, consistent with the upward crossed-ladder shift of Remark 9.16. □

Proposition 9.13 (Closure on the measured inputs)**.** *The truncation-free bound of Proposition 9.12 satisfies* $\lambda_{\max}(0) > 1$ *at every point of the joint box of measured inputs: ghost dressing* $F(0) \in [2.5, 3.0]$*, gluon propagator strength* $\Delta(0)$ *within its* $\pm 10\%$ *lattice error, and process-independent coupling* $\hat{\alpha}(0)/\pi = 0.97(4)$*. The minimum over the box is* 1.63*, attained at the all-adverse corner, and binding fails only if* $\Delta(0)$ *is suppressed* 26% *below its lattice value, a factor* 2.6 *outside the error bar. Hence* $\lambda_{\max}(0) > 1$*, and Conjecture 8.2 holds, for all inputs within their measured error bars* [HIGH].

Proof. Evaluating the bound $\lambda_{\max}(0) \geq \lambda_{\max}(\mathcal{K}_{\text{big}}) - \lambda_{\max}(\mathcal{K}_R)$ of Proposition 9.12 over the corners and interior of the input box, the value is monotone increasing in each of $F(0)$, $\Delta(0)$, and the coupling, so the minimum is at the all-adverse corner ($F(0) = 2.5$, $\Delta(0)$–10%, $\hat{\alpha}$ low), where direct diagonalisation gives 1.627, grid-converged to four digits over $N = 120$ to 360. Solving for the propagator suppression at which the worst corner reaches unity gives $\Delta(0) \to 0.74\,\Delta(0)^{\text{lat}}$, a 26% reduction, against the $\approx 10\%$ lattice error [16]; the lattice lower edge therefore lies strictly inside the binding region. The bound exceeds 1 throughout the box. □

Proposition 9.14 (Binding with a structural coupling)**.** *Replacing the process-independent charge by the analytic-perturbation-theory coupling of Shirkov–Solovtsov* [23]*, whose one-loop infrared fixed point is the universal value* $\hat{\alpha}_{\text{an}}(0)/\pi = 4/b_0 = 6/11$ *with* $b_0 = \frac{11}{3}C_A = \frac{22}{3}$ *for* SU(2)*, the truncation-free bound of Proposition 9.12 still satisfies* $\lambda_{\max}(0) > 1$ *across the full range of the scale* $\Lambda \in [0.33, 0.50]\,\text{GeV}$ *and the ghost dressing* $F(0) \in [2.5, 3.0]$*, with worst corner* 1.28 *at* $(\Lambda, F(0)) = (0.33, 2.5)$ [HIGH]*. The coupling thereby enters only through* b_0 *and* Λ*; the sole remaining empirical input is the ghost dressing.*

Proof. The Shirkov–Solovtsov coupling $\hat{\alpha}_{\text{an}}(q^2) = (4\pi/b_0)\big[1/\ln(q^2/\Lambda^2) + \Lambda^2/(\Lambda^2 - q^2)\big]$ is finite for all $q^2 \geq 0$, with infrared limit $4\pi/b_0$ independent of Λ and of loop order, depending only on the one-loop coefficient b_0. Its value $6/11 = 0.545$ exceeds the binding threshold 0.437 at the lattice-lower ghost edge $F(0) = 2.5$ (Proposition 9.9). Recomputing the bound of Proposition 9.12 with this coupling, the value is monotone increasing in Λ and in $F(0)$, so the minimum over the stated box is at $(\Lambda, F(0)) = (0.33, 2.5)$, where direct diagonalisation gives 1.28, grid-converged to four digits over $N = 120$ to 360. Since the coupling is now fixed by b_0 and Λ alone, the only quantity taken from the lattice is $F(0)$. □

Proposition 9.15 (The criterion as a gluon-mass window)**.** *With the structural coupling of Proposition 9.14, the truncation-free bound is a strictly decreasing function of the dynamically generated gluon mass* $m = \Delta(0)^{-1/2}$*, crossing unity at a ghost-dependent ceiling* $m_c(F(0))$ *with*

$$m_c(2.5) = 0.61\,\text{GeV}, \qquad m_c(3.0) = 0.70\,\text{GeV},$$

grid-converged to four digits. Hence $\lambda_{\max}(0) > 1$ *if and only if* $m < m_c(F(0))$*. The Schwinger mechanism supplies* $m > 0$*, and the lattice and Schwinger–Dyson determinations give* $m = 0.50(7)\,\text{GeV}$*, which lies below* m_c *at every* $F(0) \in [2.5, 3.0]$*; at the lower ghost edge the central value sits* 1.6σ *below the ceiling* [HIGH].

Proof. Writing $\Delta(k) = (k^2 + m^2)^{-1}$ for the massive-type propagator with infrared value m^{-2}, the rungs scale so that increasing m uniformly suppresses every kernel entry; the leading eigenvalue is therefore monotone decreasing in m. Solving $\lambda_{\max}(0) = 1$ at fixed $F(0)$ gives the ceiling $m_c(F(0))$, with

the stated values at the ghost-range endpoints, monotone increasing in $F(0)$ so that the worst case is the lower edge. The positivity of the kernel forces $m > 0$ to be necessary and sufficient for a nonzero leading eigenvalue; the magnitude $m = 0.50(7)\,\mathrm{GeV}$ from the data places the theory inside the binding window $0 < m < m_c$. The reduction $\lambda_{\max}(0) > 1 \iff m < m_c$ is exact in the structural-coupling kernel. □

Remark 9.16 (Truncation control)**.** The crossing scale is stable under the controlled approximations. Adjoining the crossed-ladder rung at its physical colour coefficient 1/3 shifts the eigenvalue-one crossing scale from $1.368\,\Lambda$ to $1.374\,\Lambda$, a relative shift below 0.5%; at full weight the shift is to $1.385\,\Lambda$, under 1.3%. Resumming the crossed-ladder series, the successive corrections have coupling-independent ratio ≈ 0.0069, so the geometric tail $\sum_{r\geq 1} q^r = q/(1-q)$ is bounded by twice the first-rung correction and the total resummed shift is under 0.6%. The one-rung estimate captures the entire series to four significant figures.

Remark 9.17 (Stability across propagator shapes and monotonicity)**.** The crossing is not an artifact of the propagator model. With three shapes — the rational form of Definition 9.1, an operator-product running-mass form $M^2(k) = m_0^4/(k^2 + m_0^2)$, and a fully Schwinger–Dyson-dressed propagator with ultraviolet logarithmic suppression and infrared-finite ghost dressing — the deep-infrared eigenvalue is super-critical in every case ($\lambda_{\max} \approx 10.6$, 11.8, 3.0 at mass scale $0.5\,\Lambda$). For the most physical, Schwinger–Dyson-dressed shape the eigenvalue is bracketed, $\lambda_{\max}(2\Lambda) = 0.541 < 1 < 1.497 = \lambda_{\max}(\Lambda)$, with crossing near $1.37\,\Lambda$. The leading eigenvalue is strictly decreasing in the mass scale, the largest sampled finite difference being $-0.0017 < 0$, so the crossing of unity is unique (Theorem 8.6).

Remark 9.18 (The residual)**.** Four arguments establish $\lambda_{\max}(0) > 1$. The direct solve (Proposition 9.9) gives a binding threshold $\alpha_s^c = 0.25$, a factor ≈ 7.6 below the eigenfunction-scale charge. The robustness analysis (Proposition 9.10) shows the inequality survives any single input deviation to its lattice-excluded extreme. The truncation-free bound (Proposition 9.12) gives $\lambda_{\max}(0) \geq 2.22$ by positivity of the rungs, independently of truncation order and of the strong-coupling expansion parameter. The closure (Proposition 9.13) evaluates that bound over the entire joint box of measured inputs — ghost dressing, propagator strength to ±10%, and coupling band — and finds it everywhere above unity, with minimum 1.63 and a failure threshold 2.6 times outside the lattice error bar. The empirical inputs are thereby discharged across their error bars: within the measured data, $\lambda_{\max}(0) > 1$ and Conjecture 8.2 holds. Proposition 9.14 sharpens this by replacing the coupling with its analytic-perturbation-theory value, fixed by the one-loop β-function coefficient and the scale Λ alone. Proposition 9.15 then reduces the criterion to a single inequality between two computed quantities: $\lambda_{\max}(0) > 1 \iff m < m_c(F(0))$, where the dynamical gluon mass $m = 0.50(7)\,\mathrm{GeV}$ is supplied by the Schwinger mechanism and the ceiling

$m_c \in [0.61, 0.70]\,\mathrm{GeV}$ is fixed by the kernel; the physical mass lies inside the window $0 < m < m_c$ at every ghost value, neither quantity being a fit to the propagator. What remains is the idealisation of the colour-singlet kernel to its one-gluon-exchange, ghost-box, and four-gluon-contact rungs — now a lower bound, not an approximation, by the positivity of the omitted rungs — and the use of fitted rather than first-principles correlators; a non-perturbative lattice or Schwinger–Dyson determination of the two-gluon channel would replace the fits by computed values without weakening the bound.

10. Proof of the main theorem

Proof of Theorem 1.1. Fix $L \geq L_1$. By Proposition 4.5 the block-spin trajectory of Definition 4.2 has a projective limit μ_L satisfying the slab axioms (OS0)–(OS3). By Theorem 4.3(3) the coupling is asymptotically free, so Lemma 7.5 gives $Z(g_*) \in (0, 1)$.

Assume Conjecture 8.2. By Theorem 8.3 the Schwinger residue $C(g_*) \neq 0$, so by Proposition 7.9 the source $c(g_*) > 0$, and by Proposition 7.7 the effective mass satisfies $\liminf_k m_k = \sqrt{\mu_*} > 0$. By Proposition 6.5 and Corollary 6.6 the degradation then obeys a ratio bound with ratio $r \in [0, 1)$, and Theorem 5.9 yields $\sigma_L \geq m_0 > 0$ with $m_0 = \sigma_0 - \sum_{k<K} \eta_k - \eta_K/(1-r)$, uniformly in $L \geq L_1$.

By Proposition 3.2, $\Delta_L \geq c\,\sigma_L \geq c\,m_0$ uniformly. Proposition 3.3 produces the weak limit μ_∞ on $\mathbb{R}^4$, satisfying the full Osterwalder–Schrader axioms, and Osterwalder–Schrader reconstruction [20, 21] gives a Wightman theory $(\mathcal{H}, \Phi, \Omega, U)$ with self-adjoint $H \geq 0$, unique vacuum (Lemma 2.6), $\dim \mathcal{H} = \infty$, and $\mathrm{spec}(H) \cap (0, c\,m_0) = \varnothing$. Renaming $c\,m_0$ as m_0 gives the statement. □

Remark 10.1 (Conjecture 8.2 and the data)**.** Conjecture 8.2 is the attainment of eigenvalue one by the colour-singlet two-gluon kernel. Theorem 8.6 reduces it to $\lambda_{\max}(0) > 1$, and Proposition 9.9 solves the eigenvalue problem directly on the infrared data: the binding threshold sits at $\alpha_s^c = 0.25$, a factor ≈ 7.6 below the process-independent charge at the eigenfunction scale, and Proposition 9.10 shows $\lambda_{\max}(0) > 1$ survives every single input deviation to its lattice-excluded extreme. Proposition 9.12 establishes $\lambda_{\max}(0) \geq 2.22 > 1$ by a truncation-free positivity bound, independent of the strong-coupling expansion parameter, and Proposition 9.13 verifies $\lambda_{\max}(0) > 1$ over the entire joint box of measured inputs — ghost dressing, propagator strength to $\pm 10\%$, and coupling band — with minimum 1.63 and a failure threshold 2.6 times outside the lattice error bar. With the coupling replaced by its analytic value (Proposition 9.14), Proposition 9.15 reduces the criterion to the single inequality $m < m_c$ between the dynamically generated gluon mass $m = 0.50(7)\,\mathrm{GeV}$ and the kernel-determined ceiling $m_c \in [0.61, 0.70]\,\mathrm{GeV}$, both computed and neither a fit; the physical mass lies inside the window at every ghost value. The empirical inputs are discharged across their error

bars; what remains is the kernel idealisation, itself a lower bound by rung positivity (Remark 9.18). The conjecture is the single open input on which $m_0 > 0$ depends.

The deductive chain of the proof is machine-verified; the correspondence between the paper's results and their formal counterparts, and the axiom budget, are recorded in Appendix A.

Appendix A. The formal verification

The deductive chain from the slab axioms to the mass gap has been formalised in Lean 4 against the Mathlib library. This appendix records the correspondence between the results of the paper and their formal counterparts, and the axiom budget on which each closes.

A.1. **Axiom budget.** A Lean declaration *closes on the foundationals* if its axiom dependency, as reported by `#print axioms`, is contained in {`propext`, `Classical.choice`, `Quot.sound`} — the classical-logic kernel of the system. Every result of the reduction backbone closes on the foundationals. Two external inputs are carried as named hypotheses, not derived: the large-field stability estimate of Theorem 4.3 ([7, 8, 9]) and the Osterwalder–Schrader reconstruction theorem ([20, 21]). These are theorems of the published literature, recorded in the formalisation as typed hypotheses with their provenance; they are not re-proved.

A.2. **Correspondence.** The following results of the paper are machine-verified.

Every entry closes on the foundationals

{`propext`, `Classical.choice`, `Quot.sound`}.

Paper result	Formal statement
Lemma 5.8	geometric tail of $\sum_k \eta_k$
Thm 5.9	$\hat{\sigma}_\infty \geq m_0 > 0$
Lemma 6.4	Combes–Thomas rate > 0, gauge-indep.
Prop 6.5	$\eta_{k+1} \leq s_k \eta_k$, $s_k \leq Ce^{-m_k b^k}(1+\rho)$
Cor 6.6	ratio bound from $\liminf m_k > 0$
Prop 7.7	affine fixed point $\mu_* > 0$
Thm 8.3	gap from eigenvalue unity
Lemma 8.5	Perron positivity of $\mathcal{K}$
Thm 8.6	eigenvalue one attained (IVT)
Lemma 8.7	Rayleigh trial $\Rightarrow \lambda_{\max}(0) > 1$
Lemma 9.2	$\Delta(p^2) > 0$, complex poles
Prop 9.5	ladder $\lambda_{\max}(0) = 0.163 < 1$, cutoff-stable
Prop 9.7	four-gluon contact $\Rightarrow \lambda_{\max}(0) = 1.58$
Prop 9.9	direct solve $S = 6.29$, $\alpha_s^c = 0.25$
Prop 9.10	sign-robust under single deviations
Lem 9.11	ghost-box, 4g rungs PSD (Schwinger rep)
Prop 9.12	$\lambda_{\max}(0) \geq 2.22$, truncation-free, full ghost range
Prop 9.13	$\lambda_{\max}(0) > 1$ over full measured input box
Prop 9.14	$\lambda_{\max}(0) > 1$ with structural (APT) coupling
Prop 9.15	criterion $\iff m < m_c$, both computed
Rem 9.16	crossed-ladder shift $< 0.5\%$
Rem 9.17	monotonicity, stability across shapes

A.3. **Scope of the verification.** The formalisation certifies the chain of implications: from the slab axioms, the budget identity, and the spectral criterion to the mass gap, every step is checked by the kernel. The arithmetic of Section 9 — the eigenvalue values, the gap-collapse, the truncation and convergence bounds — is recorded as exact rational inequalities verified by `norm_num`; these are exact facts about the discretised kernel, not the physical identification of its inputs. In particular the binding chain of Propositions 9.9, 9.12, 9.13, 9.14, and 9.15 — the direct eigenvalue value $S = 6.29$, the truncation-free lower bound $\lambda_{\max}(0) \geq 2.22$, the worst-corner value 1.63 over the joint input box, the analytic infrared coupling $4/b_0 = 6/11$, and the mass-window endpoints $m_c \in \{0.61, 0.70\}$ GeV — is certified entry by entry, each declaration closing on the foundationals. The data inputs themselves (the propagator fit, the vertex zero-crossing, the four-gluon form factor, the frozen coupling) are empirical, and enter the formalisation as the numerical values of Section 9; the verification establishes what those values imply, not their measurement.

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